

Lab 3: SQL Subqueries

Learning Objectives:

- Understand the concept of subqueries
- Write queries using subqueries in SQL

Dataset:

Use the following tables from a fictional bookstore database:

Books

- `book_id`
- `title`
- `author`
- `genre`
- `price`
- `rating`

Sales

- `sale_id`
- `book_id`
- `sale_date`
- `quantity`

Exercises:

1. Simple Subquery

Find all books priced above the average price.

2. IN Subquery:

Find titles of books authored by authors who have published more than 2 books.

3. NOT IN Subquery:

Find titles of books that have never been sold.

4. Subquery with Aggregation:

List authors whose books have an average rating above the average rating of all books.

5. Subquery with Comparison Operator:

Retrieve the details of books priced higher than the average price of all books.

6. Nested Subquery:

List the books with a price greater than the average price of books published by authors who have published exactly 1 book.

7. Subquery with Date Condition:

Find titles of books that have been sold after the date when the most expensive book was first sold.

8. Correlated Subquery:

Find each book's title along with the quantity sold in the most recent sale of that book.

9. EXISTS Subquery:

Find authors who have at least one book sold.

10. NOT EXISTS Subquery:

Find authors who have no books sold.

Deliverables:

- Submit your SQL queries as a `.sql` file.

Notes:

- Clearly comment your SQL code to explain each step.